BAS-ACC-54: Arista Access Switch Hardware/Environmental Fault SOP

Node: BAS-ACC-54 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_7156

Description

The Access Switch in Basavanagudi POP, utilizing an Arista platform, has encountered a Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss). This failure domain arises on the Access role due to the inherent architecture of modern Arista switches, which integrate a centralized ASIC (Application-Specific Integrated Circuit) responsible for packet forwarding, switching, and routing. The ASIC's memory parity errors and line card crashes, triggered by environmental factors or component degradation, compromise the system's ability to maintain network connectivity and services. This SOP aims to provide a step-by-step procedure for diagnosing and resolving Hardware/Environmental Faults on the Arista Access Switch, ensuring minimal disruption to network operations and optimal service restoration.

Pre-requisite Checks

- pre_requisite: Verify that the Basavanagudi POP is accessible, and the switch is power-cycled to ensure a clean start.

- pre_requisite: Check for redundant link status and ensure that the primary link is fully operational before initiating troubleshooting.

- pre_requisite: Confirm that the Arista Access Switch is configured for remote access, allowing for CLI connectivity and command execution.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system health, Expected Output: System health status: NORMAL, Description: This command displays the overall system health status, providing an initial assessment of the switch's operational state.

Command: show hardware cpu, Expected Output: CPU utilization: 0.00%, Description: This command monitors CPU utilization, helping to identify potential bottlenecks or overloads caused by the hardware fault.
```

```
Command: show interface all, Expected Output: Interface status:
UP/DOWN, Description: This command displays the status of all
interfaces, including the presence of any errors or alarms, which can
aid in pinpointing the affected component.
```

Resolution Steps

Step 1: Step: Power-cycle the switch to reset the system and clear any temporary errors., Command: reload now, Description: Rebooting the switch may resolve the issue by clearing stuck or corrupted packets and state information.

Step 2: Step: Verify and replace the affected line card, if necessary., Command: swapon linecard 1, Description: If the line card is faulty, replace it with a spare and reconfigure the switch to utilize the new line card.

Step 3: Step: Check and address power supply redundancy issues, if applicable., Command: power supply redundancy check, Description: Verify that the power supply redundancy is functioning correctly and, if necessary, replace the faulty power supply with a spare.

Post-Check Verification

```
Command: show system health, Expected Output: System health status:
NORMAL, Description: Verify that the system health status has returned
to normal after troubleshooting and resolution.

Command: show hardware cpu, Expected Output: CPU utilization: 0.00%,
Description: Monitor CPU utilization to ensure that the hardware fault
has not caused any long-term performance degradation.

Command: show interface all, Expected Output: Interface status:
UP/DOWN, Description: Verify that the interface status has returned to
normal, indicating that the issue has been fully resolved.
```

Roll-back Procedures

- Step: Restore the original configuration and settings., Command: copy startup-config running-config, Description: Revert the switch to its previous configuration to ensure that any unintended changes are reversed.

- Step: Reboot the switch to clear any remaining issues., Command: reload now, Description: Reboot the switch to clear any residual effects from the previous troubleshooting and resolution steps.